



Assignment No. 3

Title:

Exploratory Data Analysis (EDA) with Python/R

Aim:

To perform **Exploratory Data Analysis (EDA)** using Python/R in order to understand the structure, characteristics, patterns, and relationships within a dataset by applying statistical summaries and visualizations.

Objective:

The main objective of this assignment is to understand the concepts and practical implementation of **Exploratory Data Analysis (EDA)** using Python or R for analyzing real-world datasets. Students will learn how to import datasets, examine their structure, and apply statistical and visualization techniques to summarize and interpret the data. The assignment focuses on identifying missing values, duplicates, outliers, and understanding data distributions and relationships between variables. Through hands-on implementation using libraries such as Pandas, NumPy, Matplotlib, Seaborn (in Python) or dplyr and ggplot2 (in R), students will gain practical experience in cleaning and preparing data for further statistical analysis or machine learning tasks. This assignment will help students develop strong analytical skills and understand the importance of data preprocessing in the data science workflow.

Software Requirements:

1. Operating System: Windows / Linux / macOS
2. Python 3.x or R Programming Language
3. Jupyter Notebook / JupyterLab / Google Colab (for Python) or RStudio (for R)
4. Required Libraries (Python): Pandas, NumPy, Matplotlib, Seaborn (Scikit-learn – optional)
5. Required Packages (R): dplyr, ggplot2, tidyr, readr (summarytools – optional)
6. Command Line / Terminal or IDE (VS Code / PyCharm for Python)

Outcome:

After completing this lab, students will be able to:

1. Import and explore datasets using Python/R tools.
2. Perform data cleaning by handling missing values, duplicates, and outliers.
3. Apply statistical summaries to understand data distribution and patterns.
4. Create meaningful visualizations (charts, graphs, plots) to analyze relationships.
5. Interpret insights from the data and prepare it for further analysis or machine learning models.



Theory:

Exploratory Data Analysis (EDA) is the initial step in the data analysis process used to understand the structure, patterns, and key characteristics of a dataset before applying advanced techniques. It helps summarize data using statistical methods and visualizations. EDA allows analysts and researchers to gain insights into the data, understand its distribution, and identify important relationships among variables.

Real-world data often contains missing values, duplicates, inconsistent entries, and outliers. EDA helps identify and handle these issues to improve data quality. By detecting such problems early in the analysis process, analysts can clean and prepare the dataset for accurate and reliable results. Proper data cleaning ensures that the dataset is suitable for further statistical analysis, data mining, or machine learning tasks.

EDA mainly includes:

1. **Univariate Analysis** – Examines a single variable to understand its distribution and summary statistics such as mean, median, and standard deviation. Common plots include histograms, bar charts, and boxplots. This analysis helps in understanding the central tendency, spread, and distribution of individual variables in the dataset.
2. **Bivariate and Multivariate Analysis** – Studies relationships between two or more variables using scatter plots, correlation matrices, pair plots, and heatmaps. These techniques help identify correlations, dependencies, and patterns among variables that may influence the outcome of a model or analysis.

Statistical measures and visualization tools (like **Pandas, Matplotlib, Seaborn in Python or ggplot2 in R**) are used to analyze and interpret the data. These tools allow analysts to compute descriptive statistics, generate visualizations, and explore the dataset interactively.

Another important aspect of EDA is **data cleaning and preprocessing**. During this stage, analysts handle missing values, remove duplicate records, correct inconsistent data entries, and treat outliers. Methods such as mean or median imputation, deletion of invalid records, or transformation techniques may be used to improve data quality.

EDA also involves **detecting patterns and trends** within the dataset. Visualization techniques help identify clusters, seasonal patterns, or unusual behaviors in the data. These insights can guide decision-making and help in selecting appropriate statistical models or machine learning algorithms.

Correlation analysis is another key part of EDA. By measuring the strength and direction of relationships between variables, analysts can determine which features are important and which variables may have little impact. Correlation matrices and heatmaps are commonly used tools for



this purpose.

EDA plays a critical role in **feature selection and feature engineering**, which are important steps in building predictive models. By understanding the relationships and distributions in the dataset, analysts can select the most relevant variables and transform them into useful features for machine learning algorithms.

Furthermore, EDA helps detect **outliers and anomalies** that may significantly affect model performance. Outliers can occur due to measurement errors, incorrect data entry, or natural variations in the data. Identifying and treating these outliers ensures more accurate and reliable analysis.

Overall, the main purpose of EDA is to understand the dataset, detect patterns and anomalies, and prepare clean data for further analysis or machine learning. It provides a strong foundation for data-driven decision-making and improves the effectiveness of predictive models and statistical analyses.

It is crucial to understand that EDA is not a rigid, linear checklist but rather an iterative and exploratory dialogue with the data. An analyst does not simply run a set of commands and receive final answers. Instead, each visualization or statistical summary raises new questions. For example, a histogram might reveal a bimodal distribution, prompting the analyst to investigate what factors are splitting the data into two distinct groups. This leads to a new bivariate analysis, perhaps coloring the data points by a categorical variable like "Region" or "Customer Segment." This cyclical process of generating hypotheses, testing them visually, and refining questions is at the heart of EDA. It is this flexible, inquisitive approach that allows analysts to uncover unexpected insights that formal modeling or automated reporting might miss, fostering a deeper, more intuitive understanding of the underlying phenomena.

While EDA is often discussed in terms of its technical tools and statistical techniques, its effectiveness is profoundly enhanced by domain knowledge. A scatter plot might show a correlation, but it takes an expert in the field to understand whether that relationship is causal, spurious, or simply a known association. For instance, an EDA of healthcare data might reveal a correlation between ice cream sales and drowning incidents. Without domain knowledge, one might draw a nonsensical conclusion. However, a public health expert would recognize that both are caused by a third, "lurking" variable: hot summer weather. Domain expertise guides the analyst toward meaningful questions, helps validate the plausibility of discovered patterns, and prevents wasted effort exploring statistically significant but practically irrelevant relationships. It transforms EDA from a purely mechanical exercise into a powerful tool for scientific discovery and business intelligence.



The visualizations and summary statistics produced during EDA are not just for the analyst's private consumption; they are powerful communication tools. EDA provides the raw material for telling a compelling data story to stakeholders. Before presenting the results of a complex machine learning model, a data scientist can use simple EDA plots to set the stage. For example, showing a clear trend line of sales over time or a bar chart comparing customer churn rates across different groups makes the problem relatable and the subsequent analysis more credible. These visuals build a shared understanding and trust with the audience. By clearly articulating the "what" and "why" of the data through EDA, an analyst can effectively frame the problem, justify the need for a solution, and ultimately make their final recommendations more persuasive and impactful.

As datasets grow in volume, velocity, and variety, the traditional approach to EDA—manually plotting every variable—becomes impractical. This has led to the development of techniques for automating and scaling the exploratory process. Tools like pandas-profiling (now ydata-profiling) and sweetviz can generate comprehensive EDA reports for an entire dataset with a single line of code, providing a high-level overview of distributions, missing values, and correlations in seconds. For truly massive datasets that cannot fit into memory, analysts must leverage sampling techniques or distributed computing frameworks like Spark, which has its own libraries for scalable data exploration (e.g., Spark's DataFrame API and visualization libraries like plotly). This scalability ensures that the foundational principles of EDA remain applicable even in the most demanding big data environments, allowing analysts to maintain an overview and identify key areas for deeper investigation without being overwhelmed by scale.

Algorithm / Steps:

1. Import the required libraries (Pandas, NumPy, Matplotlib, Seaborn / dplyr, ggplot2).
2. Load the dataset into the environment (CSV/Excel file).
3. Explore the dataset structure using head(), info(), and shape functions.
4. Check and handle missing values and duplicate records.
5. Perform statistical summary analysis (mean, median, standard deviation).
6. Detect outliers using boxplots or statistical methods.
7. Perform univariate and bivariate analysis using appropriate visualizations.
8. Analyze correlations and relationships between variables.
9. Interpret the results and prepare the cleaned dataset for further analysis or modeling.

Source Code:

```
# 1. Import Libraries
import pandas as pd
import matplotlib.pyplot as plt
```



```
import seaborn as sns
```

```
# 2. Load Dataset
```

```
df = pd.read_csv("data.csv") # Replace with your file name
```

```
# 3. View Data
```

```
df.head()
```

```
# 4. Check Dataset Info
```

```
df.info()
```

```
df.describe()
```

```
# 5. Check Missing Values
```

```
df.isnull().sum()
```

```
# 6. Remove Duplicates
```

```
df = df.drop_duplicates()
```

```
# 7. Histogram (Univariate Analysis)
```

```
plt.hist(df['Age']) # Replace 'Age' with your column
```

```
plt.title("Age Distribution")
```

```
plt.show()
```

```
# 8. Correlation
```

```
df.corr()
```

Conclusion:

This experiment demonstrates the practical implementation of Exploratory Data Analysis (EDA) using Python/R for analyzing real-world datasets. By importing and exploring the dataset, we understood its structure, data types, and key statistical properties. Performing data cleaning helped in handling missing values, duplicates, and outliers effectively. Through visualizations and correlation analysis, meaningful patterns and relationships between variables were identified. This experiment highlights the importance of EDA as a crucial step in the data science process, ensuring that data is clean, consistent, and ready for further analysis or machine learning modeling.

The hands-on approach allowed us to apply various data manipulation techniques using libraries such as Pandas, Matplotlib, and Seaborn in Python (or ggplot2 and dplyr in R). Univariate analysis through histograms and box plots provided insights into individual variable distributions and helped detect outliers. Bivariate analysis using scatter plots and correlation matrices revealed relationships between variables, guiding feature selection for predictive modeling. The visualizations not only made complex data patterns easier to understand but also helped in communicating findings effectively. Overall, this experiment reinforces that thorough EDA is



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essential for building reliable and accurate data-driven solutions.

Questions:

1. What is Exploratory Data Analysis (EDA) and why is it important?
2. What are the different types of analysis performed in EDA?
3. How can missing values and outliers be handled during EDA?
4. Explain the difference between univariate and bivariate analysis with examples.
5. Why is data visualization important in Exploratory Data Analysis?

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